Practice Set 11 Paper 2 – Urology

Contents

1. Q1. Management of metastatic RCC (LAQ, 25)
2. Q2. Urinary crystals (LAQ, 25)
3. Q3a. Biological effects of cryosurgery (SAQ, 10)
4. Q3b. Mitomycin (SAQ, 10)
5. Q3c. HOTA and its modification (SAQ, 10)
6. Q3d. Marginal kidney donor (SAQ, 10)
7. Q3e. Medical expulsive Rx for urinary calculi. (SAQ, 10)
8. Q3f. Prostatectomy incontinence (SAQ, 10)

Practice paper compiled from the question bank.

Q1. Management of metastatic RCC

Type: LAQ (25 marks) (also answered in 2020-Summer Paper 2, Q1)

Introduction

- About one-third of RCC (renal cell carcinoma) patients have metastases at diagnosis, and 20–40% of localised tumours metastasise later.
- Treatment moved from cytokines to VEGF (vascular endothelial growth factor)-targeted TKIs (tyrosine kinase inhibitors), and now to ICI (immune checkpoint inhibitor) combinations. Surgery has become selective.

Assessment before treatment

- **Clinical:** symptoms of metastases (cough, bone pain, neurological deficit), paraneoplastic syndromes (hypercalcaemia, polycythaemia, Stauffer syndrome), performance status (Karnofsky or ECOG [Eastern Cooperative Oncology Group]).
- **Laboratory:** haemoglobin, corrected calcium, neutrophils, platelets, LDH (lactate dehydrogenase), creatinine, liver tests.
- **Imaging:** contrast CT of chest, abdomen and pelvis. Bone scan or brain MRI only when symptoms or tests suggest spread.
- **Histology:** nephrectomy or **percutaneous biopsy** before systemic therapy. It separates clear cell from non-clear cell RCC and shows sarcomatoid change.
- **Prognostic risk group** (decides first-line therapy):

Adverse factor	MSKCC (Memorial Sloan-Kettering), cytokine era	IMDC (International mRCC Database Consortium), TKI era
Karnofsky <80%	Yes	Yes
Haemoglobin < lower limit of normal	Yes	Yes
Corrected calcium raised	Yes	Yes
LDH >1.5 × upper limit of normal	Yes	No
Time factor	No prior nephrectomy	<1 year from diagnosis to systemic therapy
Neutrophils, platelets raised	No	Yes (two factors)

- Favourable 0 factors, intermediate 1–2, poor ≥3. IMDC median survival: 43.2, 22.5 and 7.8 months.

1. First-line systemic therapy, clear cell RCC

IMDC risk	Preferred	Alternatives
Favourable	Pembrolizumab + axitinib; nivolumab + cabozantinib; pembrolizumab + lenvatinib	Sunitinib or pazopanib; active surveillance for slow, low-volume disease
Intermediate/poor	The same three, **or nivolumab + ipilimumab**	Cabozantinib; sunitinib; pazopanib

- **Rationale:** VHL (von Hippel–Lindau) loss lets HIF (hypoxia-inducible factor) accumulate and drive VEGF, the TKI target. ICIs block PD-1 (programmed death-1) or CTLA-4 (cytotoxic T-lymphocyte antigen-4), releasing the brake on T cells and giving durable responses.

- **Trials (all versus sunitinib):** CheckMate 214 (nivolumab + ipilimumab; overall survival benefit in intermediate/poor risk); KEYNOTE-426 (pembrolizumab + axitinib); CheckMate 9ER (nivolumab + cabozantinib); CLEAR (pembrolizumab + lenvatinib). Benefit is independent of PD-L1 (programmed death ligand-1) status.

2. Later lines: switch the mechanism

- After an IO (immuno-oncology) combination: a VEGF TKI not used before (cabozantinib, axitinib, tivozanib).
- After both IO and TKI: **belzutifan** (HIF-2α inhibitor); LITESPARK-005 showed better PFS (progression-free survival) than everolimus.
- After TKI alone: nivolumab (CheckMate 025) or cabozantinib (METEOR), both with survival gain.

3. Cytoreductive nephrectomy (CN): from routine to selective

- Cytokine era: SWOG 8949 and EORTC (European Organisation for Research and Treatment of Cancer) trials showed longer survival with CN + interferon-α.
- **CARMENA (2018):** sunitinib alone was non-inferior to CN + sunitinib in MSKCC intermediate/poor risk (median survival 18.4 vs 13.9 months).
- **SURTIME:** deferred CN after sunitinib was safe; survival favoured deferral.
- **Now:** immediate CN for fit patients with low metastatic burden who do not need immediate systemic therapy; **deferred CN** in responders. Avoid in poor risk or poor performance status.
- Palliative nephrectomy or angioembolisation for haematuria or pain.

4. Metastasectomy and local therapy

- Complete resection of solitary or few metastases gives 5-year survival of 35–50%. Better with complete resection, solitary lung lesion, disease-free interval >1 year and metachronous disease.
- **SBRT (stereotactic body radiotherapy)**, radiosurgery for brain metastases, or ablation for oligometastatic disease.
- Palliative radiotherapy or surgery for cord compression and weight-bearing bones.

5. Adjuvant therapy after complete resection

- **KEYNOTE-564:** 1 year of pembrolizumab after surgery in high-risk clear cell RCC, including **M1 NED** (no evidence of disease after metastasectomy). Disease-free survival improved; later report showed overall survival benefit.

6. Non-clear cell RCC and supportive care

- Non-clear cell: clinical trial preferred; cabozantinib or sunitinib; gemcitabine + platinum for collecting duct carcinoma.
- Bone metastases: zoledronic acid or denosumab.
- Immune-related adverse events (colitis, hepatitis, hypophysitis, thyroiditis) need steroids and drug interruption.
- **Also know:** high-dose IL-2 (interleukin-2) gives durable complete response in a few percent; temsirolimus for poor risk; chemotherapy and hormones are ineffective.

Follow-up

- CT every 8–12 weeks on therapy (response by RECIST [Response Evaluation Criteria in Solid Tumours]); monitor thyroid, liver, blood pressure and cortisol.

Recent advances

- Belzutifan, also approved for VHL-associated tumours; biomarkers to choose IO/IO or IO/TKI; neoadjuvant ICI trials.

```
Metastatic RCC -> biopsy + IMDC risk
   -> IO/TKI (all risks) or nivo + ipi (intermediate/poor)
   -> deferred CN / metastasectomy / SBRT in responders
   -> progression: switch class (TKI, belzutifan)
```

Diagram to draw: the flowchart above.

Key points to remember:

- IMDC: Karnofsky, haemoglobin, calcium, neutrophils, platelets, <1 year to therapy; poor = ≥3.
- IO/TKI for all risk groups; nivolumab + ipilimumab for intermediate/poor only.
- CARMENA and SURTIME: no routine upfront CN.
- Adjuvant pembrolizumab (KEYNOTE-564), including M1 NED.

Q2. Urinary crystals

Type: LAQ (25 marks)

Introduction

- Crystals are precipitated urinary solutes seen on microscopy of the sediment. They occur in both normal and diseased states.
- Their value lies in suggesting stone type, metabolic disease, infection or drug toxicity. **Cystine is the one crystal that is never normal.**

How crystals form (physicochemistry)

- Urine is normally **supersaturated** with calcium oxalate and phosphate. Crystals form only when the concentration product passes a critical level.

Saturation zone	What happens
Undersaturated (below Ksp, solubility product)	No crystals; stones may dissolve
Metastable (Ksp to Kf, formation product)	No spontaneous nucleation; growth on existing crystals; inhibitors effective
Unstable (above Kf)	Spontaneous nucleation, growth and aggregation

- **Nucleation** in vivo is mostly heterogeneous (on cell debris or other crystals). Crystals then grow, aggregate and are retained, for example on **Randall plaque**.
- **Inhibitors:** citrate, magnesium, pyrophosphate, Tamm–Horsfall protein, nephrocalcin, osteopontin. No inhibitor exists for uric acid or cystine, so these depend on pH and concentration.
- **Determinants of crystalluria:** urine volume, solute excretion, pH, temperature and inhibitors.

Examination of urine for crystals

- Fresh midstream specimen, examined within 2 hours at body or room temperature. **Refrigeration precipitates crystals** that may not redissolve.
- Dipstick pH (physiological range 4.5–8.0); centrifuged sediment; polarised light helps identify crystals.
- Phosphaturia (cloudy alkaline urine) **clears with acetic acid**; pyuria does not.
- Candida mimics calcium oxalate and red cells; budding and hyphae separate it.

Classification by urine pH and shape

Urine pH	Crystal	Shape	Clinical significance
Acidic	**Uric acid**	Rhomboid or lemon-shaped plates, rosettes; yellow-brown	Uric acid stones (pH <5.5; pKa 5.35), gout, tumour lysis
Acidic	Amorphous urates	Pink "brick-dust" granules	Usually normal (concentrated, cooled urine)
Acidic	**Cystine**	**Hexagonal** colourless plates	Always abnormal: cystinuria
Acidic or alkaline	**Calcium oxalate dihydrate**	**Envelope** (octahedral)	Common, often normal; hypercalciuria
Acidic or alkaline	Calcium oxalate monohydrate	Dumbbell or oval	Hyperoxaluria; ethylene glycol poisoning

Urine pH	Crystal	Shape	Clinical significance
Alkaline	**Triple phosphate (struvite)**	**Coffin-lid** prisms	Urease-producing infection (Proteus); staghorn stones
Alkaline	Amorphous phosphate	White granules	Benign phosphaturia
Alkaline	Calcium phosphate (brushite, apatite)	Needles or rosettes; amorphous	RTA (renal tubular acidosis), hyperparathyroidism
Alkaline	Ammonium biurate	"Thorny apple" spheres	Old specimens; ammonium urate stones (diarrhoea, laxative abuse, ileostomy)
Any	Cholesterol	Notched flat plates	Nephrotic syndrome, lipiduria

- **Rare metabolic crystals:** 2,8-dihydroxyadenine, brown spiculated spheres (APRT [adenine phosphoribosyltransferase] deficiency, radiolucent stones); xanthine (hereditary xanthinuria, high-dose allopurinol); tyrosine needles and leucine spheres (severe liver disease).

Drug crystals

- **Two routes:** the drug itself precipitates (direct), or it changes urine chemistry (indirect).

Drug	Crystal and features	Prevention
Indinavir (protease inhibitor)	Needles or fan-shaped plates; crystalluria in about two-thirds of new users; stones 4–13%; radiolucent, often invisible on CT	Hydration; more soluble below pH 5.5
Atazanavir	Highest stone risk of newer antiretrovirals	Hydration
Sulfonamides (sulfadiazine)	"Sheaves of wheat"	Hydration, alkalinise
Acyclovir (high-dose intravenous)	Birefringent needles; acute kidney injury	Slow infusion, hydration
Amoxicillin, ciprofloxacin (high dose)	Needles; worse in alkaline urine	Hydration
Methotrexate (high dose)	Crystal nephropathy	Alkalinise, hydration
Triamterene, ephedrine, guaifenesin, silicate antacids	Stones of drug or metabolite	Stop the drug

- **Indirect:** loop diuretics and steroids (hypercalciuria, calcium stones); acetazolamide and topiramate (alkaline urine, hypocitraturia, calcium phosphate); cytotoxics (uric acid); laxative abuse (ammonium urate).

Clinical significance

1. **Diagnosis of stone type** before stone analysis is available; crystals in a stone former help direct metabolic work-up.
2. **Recurrence risk:** persistent crystalluria in repeated early-morning samples, and large or aggregated crystals, suggest active stone disease.
3. **Infection:** struvite crystals with urine pH >7.2 signal a urease-producing organism.
4. **Acute kidney injury:** uric acid (tumour lysis), calcium oxalate monohydrate (ethylene glycol) and drug crystals cause intratubular obstruction.
5. **Monitoring therapy:** absence of cystine or uric acid crystals confirms adequate alkalinisation.

- Most crystalluria in healthy people is transient and harmless.

Management principles

- Repeat on a fresh, warm specimen; confirm pH; 24-hour urine metabolic evaluation and stone analysis.
- **Fluids** for urine output above 2–2.5 L/day.
- **pH manipulation:** alkalinise (potassium citrate) for uric acid, cystine, methotrexate and sulfonamides; avoid alkali in calcium phosphate and struvite.
- Treat the cause: antibiotics and stone clearance for struvite; stop or change the drug; specific therapy (thiols for cystine, allopurinol for uric acid).

Diagram to draw: sketches of hexagon (cystine), envelope (calcium oxalate), coffin lid (struvite), rhomboid (uric acid), sheaf of wheat (sulfonamide).

Key points to remember:

- Cystine hexagons are always pathological.
- Acidic: uric acid, cystine; alkaline: struvite, phosphates; calcium oxalate at any pH.
- Indinavir: crystalluria in two-thirds, stones 4–13%, radiolucent.
- Refrigerated urine precipitates crystals: examine fresh within 2 hours.

Q3a. Biological effects of cryosurgery

Type: SAQ (10 marks) (related: cryotherapy for carcinoma prostate, 2015-Summer Paper 4, Q3f)

Principle

- Tissue destruction by extreme cold. **Argon** expanding in the probe tip cools by the **Joule–Thomson effect**; **helium** expanding warms (active thaw).
- Used for small renal masses and prostate cancer.

Mechanisms of cell death

Tissue dies during **both freezing and thawing**, by four mechanisms:

Mechanism	What happens
1. Direct cell injury	*Freeze:* extracellular water freezes first and becomes hyperosmolar; cells dehydrate and membranes distort. Rapid cooling forms **intracellular ice**, which ruptures organelles. *Thaw:* water rushes back, cells swell and burst; ice recrystallises into larger crystals
2. Vascular injury	Endothelial damage, platelet activation and **thrombosis** cause ischaemic necrosis; the main delayed killing effect
3. Apoptosis	Sublethal cold at the periphery triggers programmed cell death
4. Immunomodulation	Necrotic cells release antigens and DAMPs (damage-associated molecular patterns); dendritic cells activate T cells against tumour

- **End result:** coagulative necrosis, then fibrous scar.

Factors deciding lethality

- **Temperature:** normal renal tissue dies at about −19.4 °C; tumour may need −40 to −50 °C. **Target ≤−40 °C.**
- **Fast freeze, slow thaw** maximise injury.
- **Double freeze–thaw cycle** enlarges the necrotic zone (standard of care); each freeze about 8–10 minutes.
- **Heat sink:** nearby large vessels warm tissue; more probes or longer freezing needed.

Zones of the ice ball

- Centre: complete necrosis. Edge: 0 °C; below −20 °C only about 3 mm inside the edge. The lethal −40 °C isotherm lies well inside the visible ice ball.
- So the ice ball must extend **5–10 mm beyond the tumour**.

Clinical relevance

- Real-time imaging shows the ice ball; thermocouples and a **urethral warmer** protect sphincter and urethra in prostate cryotherapy.
- **Also know:** passive versus active thaw (both used); insulated probe shafts reduce skin injury.

Diagram to draw: ice ball with probe, −40 °C isotherm inside the 0 °C edge and tumour margin.

Key points to remember:

- Argon freezes, helium thaws (Joule–Thomson).
- Four mechanisms: direct injury, vascular thrombosis, apoptosis, immune effect.
- ≤−40 °C, two freeze–thaw cycles, margin 5–10 mm.

Q3b. Mitomycin

Type: SAQ (10 marks)

Pharmacology

- MMC (mitomycin C) is an antitumour antibiotic from *Streptomyces caespitosus*.
- **Mechanism:** after reductive activation in the cell it acts as an **alkylating agent**, cross-linking DNA and stopping DNA synthesis; cell-cycle non-specific.
- Molecular weight 334 Da: **little systemic absorption** from the bladder.

Uses in NMIBC (non-muscle-invasive bladder cancer)

1. **Single immediate post-TURBT (transurethral resection of bladder tumour) instillation:** 40 mg. Given **within 6 hours** it reduces recurrence; at 24 hours it does not, because it works by preventing implantation of floating cells. Best for a solitary low-grade papillary tumour. Avoid if perforation or extensive resection is suspected.
2. **Adjuvant induction:** 40 mg weekly for 6 weeks (± maintenance) in low- and intermediate-risk disease. Reduces recurrence but **not progression**; BCG (bacille Calmette–Guérin) with maintenance is superior in high risk.
3. **Optimised delivery (Au, 2001):** empty bladder, overnight fluid restriction, sodium bicarbonate (less drug breakdown), 40 mg in 20 mL; retain 1–2 hours.
4. **Device-assisted:** EMDA (electromotive drug administration) increases tissue uptake; chemohyperthermia at 42 °C (RITE [radiofrequency-induced thermochemotherapeutic effect], HIVEC [hyperthermic intravesical chemotherapy]).
5. **Sequential gemcitabine then MMC** for BCG-exposed disease.

Other urological uses

- **Upper tract:** single bladder dose after nephroureterectomy (ODMIT-C: bladder recurrence 28% to 16% at 1 year). **UGN-101 (mitomycin gel)**, FDA (Food and Drug Administration) approved 2020 for low-grade UTUC (upper tract urothelial carcinoma): complete response 59%, ureteric stricture about 44%. UGN-102 for low-grade intermediate-risk bladder tumours.
- **Bladder-sparing chemoradiation:** 5-FU (5-fluorouracil) + MMC (BC2001: better locoregional control than radiation alone).
- **Antifibrotic:** injected after incision of recurrent bladder neck contracture or urethral stricture.

Side effects

- Chemical cystitis (30–40%); **skin rash, palmar or genital dermatitis** (signature toxicity; wash after voiding); contracted bladder (about 5%); encrustation cystitis; myelosuppression rare.
- Extravasation through a perforation can cause pelvic necrosis.

Diagram to draw: timeline: TURBT, single MMC within 6 hours, then weekly induction ×6.

> **Key points to remember:**
> - Alkylating antibiotic; 40 mg; single dose within 6 hours of TURBT.
> - Reduces recurrence, not progression.
> - Signature toxicity: skin rash; contracted bladder about 5%.

Q3c. HOTA and its modification

Type: SAQ (10 marks)

Assumed meaning: India's Transplantation of Human Organs Act (THOA) 1994, its 2011 amendment and 2014 Rules.

THOA 1994: aims

- Regulates removal, storage and transplantation of organs and **prevents commercial dealing**.
- **Recognised brain-stem death as legal death**, allowing deceased donation.

Main provisions (1994)

- **Living donors: near relatives** (spouse, parents, children, siblings) after verification. Others, donating "by reason of affection or attachment", need **Authorisation Committee** approval.
- **Appropriate Authority** registers and inspects hospitals; only registered centres may transplant.
- **Brain-stem death certification by a Board of four:** doctor in charge of the hospital; independent specialist from an approved panel; neurologist or neurosurgeon; treating doctor. **None from the transplant team.** Two sets of tests ≥6 hours apart.
- **Organ sale is a punishable offence.**

2011 amendment (THOTA, Transplantation of Human Organs and Tissues Act)

Change	Purpose
Tissues included	Regulates tissue banks
Grandparents and grandchildren become near relatives	Wider related donor pool
Swap (paired) donation between near-relative pairs	Overcomes ABO or crossmatch incompatibility
Board: if no neurologist, a nominated surgeon, physician, anaesthetist or intensivist	Smaller hospitals can certify
Transplant coordinator mandatory	Organises donation
Required request: ICU (intensive care unit) staff must ask families of brain-dead patients	More deceased donors
National network and registry (NOTTO [National Organ and Tissue Transplant Organisation])	Fair allocation
Higher penalties for organ trade (up to 10 years' imprisonment; fine up to ₹1 crore)	Deterrence

Rules 2014 (THOT Rules)

- Standard forms for consent and certification (**Form 10** for brain-stem death); brainstem reflexes and apnoea test set out.
- Authorisation Committee interviews donor and recipient to exclude coercion or payment.
- **Also know:** regional and state networks (ROTTO, SOTTO); 2023 guidelines removed the domicile rule and age limit for deceased-donor listing.

Diagram to draw: flowchart: brain-dead patient, Board of four, family consent, NOTTO allocation.

Key points to remember:

- THOA 1994: brain-stem death legal; anti-commercial; Authorisation Committee for unrelated donors.
- 2011: tissues, swap, grandparents, coordinator, required request, higher penalties.
- Rules 2014: Form 10; Board of four; tests ≥6 hours apart.

Q3d. Marginal kidney donor

Type: SAQ (10 marks) (also answered in 2024-Winter Paper 4, Q3e and 2016-Summer Paper 3, Q3f)

Definition

- A **marginal (extended) donor** is one whose kidney carries **more DGF (delayed graft function) and shorter graft survival** than a standard donor. Such kidneys are used because of organ shortage.
- They still give better survival than staying on dialysis (Ojo, 2001).

Deceased marginal donors

- **ECD (expanded criteria donor), UNOS (United Network for Organ Sharing):** age **60 or over**, or age **50–59 with 2 or more of:** hypertension, **creatinine >1.5 mg/dL**, death from **CVA (cerebrovascular accident)**. Graft-loss risk about 1.7 times a standard donor.
- **KDPI (Kidney Donor Profile Index) >85%** replaced ECD in the US (2014); it uses 10 donor factors. Longevity matching pairs low-KDPI kidneys with recipients with the best EPTS (Estimated Post-Transplant Survival).
- **DCD (donation after circulatory death):** warm ischaemia raises DGF; long-term survival similar to DBD (donation after brain death).
- **Others:** donor AKI (acute kidney injury); HCV (hepatitis C virus)-positive donors with antiviral treatment; paediatric donors under 15 kg (en bloc).

Living marginal donors

- Older donors, controlled hypertension, obesity, small stones, multiple renal arteries. Acceptable after careful evaluation and consent.

Evaluation: Remuzzi biopsy score

- Pre-implantation biopsy; four compartments, each 0–3 (total 0–12):

Component	0	1	2	3
Glomerular sclerosis	None	<20%	20–50%	>50%
Tubular atrophy	None	<20%	20–50%	>50%
Interstitial fibrosis	None	<20%	20–50%	>50%
Arterial narrowing	None	Wall < lumen	Wall = or slightly > lumen	Severe narrowing

- **0–3: single** transplant; **4–6: dual** kidney transplant; **7–12: discard**.

Optimising use

- Short cold ischaemia; **HMP (hypothermic machine perfusion)** reduces DGF without changing long-term survival.
- **Old-for-old** allocation (Eurotransplant Senior Program, 65 or over).
- Informed consent; induction; delayed calcineurin inhibitor if DGF.
- **Urological point:** older kidneys have more urine leak and ureteric stricture.

Diagram to draw: Remuzzi grid with four compartments scored 0–3.

> **Key points to remember:**
> - ECD: ≥60, or 50–59 with 2 of hypertension, creatinine >1.5, CVA death.
> - KDPI >85% replaced ECD (2014).
> - Remuzzi 0–3 single, 4–6 dual, ≥7 discard.

Q3e. Medical expulsive Rx for urinary calculi.

Type: SAQ (10 marks) (also answered in 2021-Summer Paper 2, Q3d)

Definition

- MET (medical expulsive therapy): drugs that help a ureteric stone pass into the bladder. **α-blockers** are the main agents.

Rationale: natural history

Width	Spontaneous passage	Location	Spontaneous passage
0–3 mm	98%	Vesicoureteric junction (small stones)	92%
5 mm	65%	Distal ureter	75%
>6.5 mm	9%	Proximal ureter	48%

Mechanism

- **α1 receptors** (mainly α1A and α1D) are abundant in the distal ureter. Blockade reduces spasm and the force and frequency of peristalsis, so a larger urine bolus builds higher pressure above the stone.
- Others: calcium-channel blockers (nifedipine); PDE5 (phosphodiesterase-5) inhibitors under study. **NSAIDs (non-steroidal anti-inflammatory drugs) relieve pain but do not aid passage.**

Drugs

- **Tamsulosin 0.4 mg daily** (most studied); **silodosin 8 mg daily** (better expulsion in meta-analyses but twice the retrograde ejaculation); alfuzosin, doxazosin.

Evidence

Study	Result
SUSPEND (Pickard, 2015): 1136 patients; tamsulosin, nifedipine or placebo	No difference in need for intervention at 4 weeks (19% vs 20%)
STONE (Meltzer, 2018): stones up to 9 mm	Tamsulosin no better than placebo
Meta-analyses; large distal-stone RCT (randomised controlled trial)	Benefit for **distal stones >5 mm**

- Stratified, the trials agree: benefit lies in **distal stones of 5–10 mm**; smaller ones pass anyway, larger rarely.

Indications (AUA [American Urological Association] and EAU [European Association of Urology])

- Ureteric stone **<10 mm**, especially distal, when immediate intervention is not needed.
- **After SWL (shock wave lithotripsy)** to clear fragments. Children: off-label but effective.

Contraindications (drain urgently instead)

- Obstruction with infection or sepsis, solitary or bilateral obstruction, renal impairment, refractory pain or vomiting.

Protocol

- NSAID analgesia; counsel on off-label use, dizziness and retrograde ejaculation.

- KUB (kidney–ureter–bladder radiograph) or ultrasound follow-up; **intervene after 4–6 weeks** if not passed.

Diagram to draw: ureter with α1 receptors concentrated distally and a stone above the vesicoureteric junction.

> **Key points to remember:**
> - Tamsulosin 0.4 mg; best for distal stones 5–10 mm.
> - SUSPEND 2015: no benefit in an unselected population.
> - Trial of passage 4–6 weeks, then intervene.

Q3f. Prostatectomy incontinence

Type: SAQ (10 marks) (also answered in 2021-Summer Paper 2, Q3a)

Definition and incidence

- PPI (post-prostatectomy incontinence) is mainly **SUI (stress urinary incontinence)** after RP (radical prostatectomy), the commonest cause of male SUI.
- Rates vary with definition and reporting; continence improves for up to 2 years.

Pathophysiology

- RP removes the internal sphincter (bladder neck and prostatic urethra), so continence depends entirely on the **rhabdosphincter**.
- Causes: sphincter or nerve injury at the apex, short membranous urethra, afferent denervation; also detrusor overactivity, low compliance and **VUAS (vesicourethral anastomotic stenosis)**.
- Nerve sparing and posterior reconstruction give **earlier, not better final** continence.

Evaluation

- History (severity, radiotherapy), bladder diary.
- **24-hour pad weight:** mild <150 g, moderate <400 g, severe more (thresholds vary).
- **Standing cough test** grades 0–4: leak reported but not seen; delayed drops; early drops; delayed stream; early persistent stream.
- Urinalysis, creatinine, PSA (prostate-specific antigen). **Cystoscopy** excludes VUAS and shows sphincter coaptation; urodynamics if mixed symptoms.

Management

1. **Wait 6–12 months.** Supervised **PFMT (pelvic floor muscle training)** with biofeedback, ideally started before surgery; pads, condom catheter or penile clamp.
2. **Duloxetine** (off-label). **Bulking agents are not advised.**
3. **Surgery** (AUA/SUFU [American Urological Association/Society of Urodynamics, Female Pelvic Medicine and Urogenital Reconstruction] 2019; EAU [European Association of Urology]):

Situation	Choice
Mild–moderate, no radiotherapy	**Male sling**: AdVance XP (repositions the bulbar urethra; needs a coapting sphincter) or adjustable ATOMS (adjustable transobturator male system)
Severe, or any radiotherapy	**AUS (artificial urinary sphincter), AMS 800**: gold standard; cuff pressure 61–70 cm H_2O
ProACT balloons	Expert centres; avoid after radiotherapy

- **MASTER trial:** male sling non-inferior to AUS at 12 months.
- AUS: social continence 55–77%; revision 10–29% at 3 years; lifespan 8–10 years. Erosion risk higher after radiotherapy; **transcorporal cuff** for fragile urethra.
- **Also know:** climacturia; treat VUAS first.

Diagram to draw: AMS 800 with cuff, scrotal pump and pressure-regulating balloon.

> **Key points to remember:**
> - Wait 6–12 months with PFMT before surgery.
> - Sling for mild–moderate; AUS for severe or irradiated.
> - MASTER: sling non-inferior to AUS at 12 months.

Rapid Revision Sheet – Practice Set 11 Paper 2

All the key points from this paper in one place. Read this in the last 30 minutes before the exam.

Q1. Management of metastatic RCC (LAQ)

- IMDC: Karnofsky, haemoglobin, calcium, neutrophils, platelets, <1 year to therapy; poor = ≥3.
- IO/TKI for all risk groups; nivolumab + ipilimumab for intermediate/poor only.
- CARMENA and SURTIME: no routine upfront CN.
- Adjuvant pembrolizumab (KEYNOTE-564), including M1 NED.

Q2. Urinary crystals (LAQ)

- Cystine hexagons are always pathological.
- Acidic: uric acid, cystine; alkaline: struvite, phosphates; calcium oxalate at any pH.
- Indinavir: crystalluria in two-thirds, stones 4–13%, radiolucent.
- Refrigerated urine precipitates crystals: examine fresh within 2 hours.

Q3a. Biological effects of cryosurgery (SAQ)

- Argon freezes, helium thaws (Joule–Thomson).
- Four mechanisms: direct injury, vascular thrombosis, apoptosis, immune effect.
- ≤−40 °C, two freeze–thaw cycles, margin 5–10 mm.

Q3b. Mitomycin (SAQ)

- Alkylating antibiotic; 40 mg; single dose within 6 hours of TURBT.
- Reduces recurrence, not progression.
- Signature toxicity: skin rash; contracted bladder about 5%.

Q3c. HOTA and its modification (SAQ)

- THOA 1994: brain-stem death legal; anti-commercial; Authorisation Committee for unrelated donors.
- 2011: tissues, swap, grandparents, coordinator, required request, higher penalties.
- Rules 2014: Form 10; Board of four; tests ≥6 hours apart.

Q3d. Marginal kidney donor (SAQ)

- ECD: ≥60, or 50–59 with 2 of hypertension, creatinine >1.5, CVA death.
- KDPI >85% replaced ECD (2014).
- Remuzzi 0–3 single, 4–6 dual, ≥7 discard.

Q3e. Medical expulsive Rx for urinary calculi. (SAQ)

- Tamsulosin 0.4 mg; best for distal stones 5–10 mm.
- SUSPEND 2015: no benefit in an unselected population.
- Trial of passage 4–6 weeks, then intervene.

Q3f. Prostatectomy incontinence (SAQ)

- Wait 6–12 months with PFMT before surgery.
- Sling for mild–moderate; AUS for severe or irradiated.
- MASTER: sling non-inferior to AUS at 12 months.